

[S A F E T E N S O R S] S a f e t e n s o r s

Safetensors documentation Safetensors and get access to the augmented documentation experience to get started Safetensors is a new simple format for storing tensors that is still fast (zero-copy) and t...

来源：<https://huggingface.co/docs/safetensors/index>

分类：12_模型部署与工程化

S a f e t e n s o r s d o c u m e n t a t i o n

S a f e t e n s o r s

a n d g e t a c c e s s t o t h e a u g m e n t e d d o c u m e n t a t i o n e x p

t o g e t s t a r t e d

S a f e t e n s o r s i s a n e w s i m p l e f o r m a t f o r s t o r i n g t e n s o r s
t h a t i s s t i l l f a s t (z e r o - c o p y) . S a f e t e n s o r s i s r e

I n s t a l l a t i o n

w i t h p i p :

p i p i n s t a l l s a f e t e n s o r s

w i t h c o n d a :

c o n d a i n s t a l l - c c o n d a - f o r g e s a f e t e n s o r s

U s a g e

L o a d t e n s o r s

f r o m s a f e t e n s o r s i m p o r t s a f e _ o p e n

t e n s o r s = { }

w i t h s a f e _ o p e n (" m o d e l . s a f e t e n s o r s " , f r a m e w o r k = " p t

f o r k i n f . k e y s () :

t e n s o r s [k] = f . g e t _ t e n s o r (k)

Loading only part of the tensors (interesting when

```
from safetensors import safe_open
```

```
tensors = {}
```

```
with safe_open("model.safetensors", framework="pt")
```

```
    tensor_slice = f.get_slice("embedding")
```

```
    vocab_size, hidden_dim = tensor_slice.get_shape()
```

```
    tensor = tensor_slice[:, :hidden_dim]
```

Save tensors

```
import torch
```

```
from safetensors.torch import save_file
```

```
tensors = {
```

```
    "embedding": torch.zeros((2, 2)),
```

```
    "attention": torch.zeros((2, 3))
```

```
}
```

```
save_file(tensors, "model.safetensors")
```

Format

Let's say you have safetensors file named model.safetensors. It has the following internal format:

```
model.safetensors
```

```
model.safetensors
```

Featured Projects

Safetensors is being used widely at leading AI enterprises like EleutherAI, and StabilityAI. Here is a non-exhaustive list of projects using safetensors:

```
huggingface/transformers
```

ml - explore / ml x

huggingface / candle

AUTOMATIC1111 / stable - diffusion - webui

Llama - cpp

microsoft / TaskMatrix

hpcaitech / ColossalAI

huggingface / pytorch - image - models

CivitAI

huggingface / diffusers

corey lowman / dfdx

invoke - ai / InvokeAI

oobabooga / text - generation - webui

Sanster / llama - cleaner

PaddlePaddle / PaddleNLP

AI GC - Audio / AudioGPT

brycedrennan / imaginAIry

comfy anonymous / ComfyUI

LianjiaTech / BELLE

alvarobartt / safejax

MaartenGr / BERTopic

rachthree / safe structures

j u s t i n c h u b y / o n n x - s a f e t e n s o r s